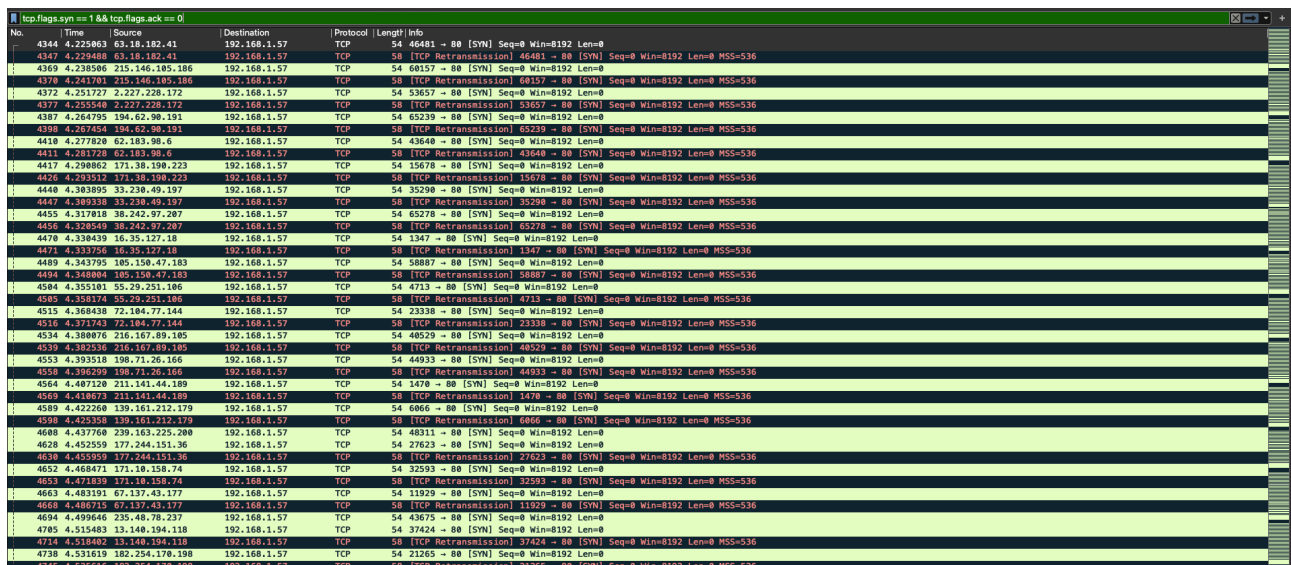


Assignment-1

Q1. SYN Flood Attack

The SYN flood simulation generated a large number of TCP SYN packets directed toward the target host. Packet analysis in Wireshark revealed multiple SYN packets originating from randomized source IP addresses. The TCP handshake was not completed because ACK packets were absent. The IO graph showed a sudden spike in SYN packet rate, indicating abnormal connection attempts. These characteristics match the typical behavior of a SYN flood attack.

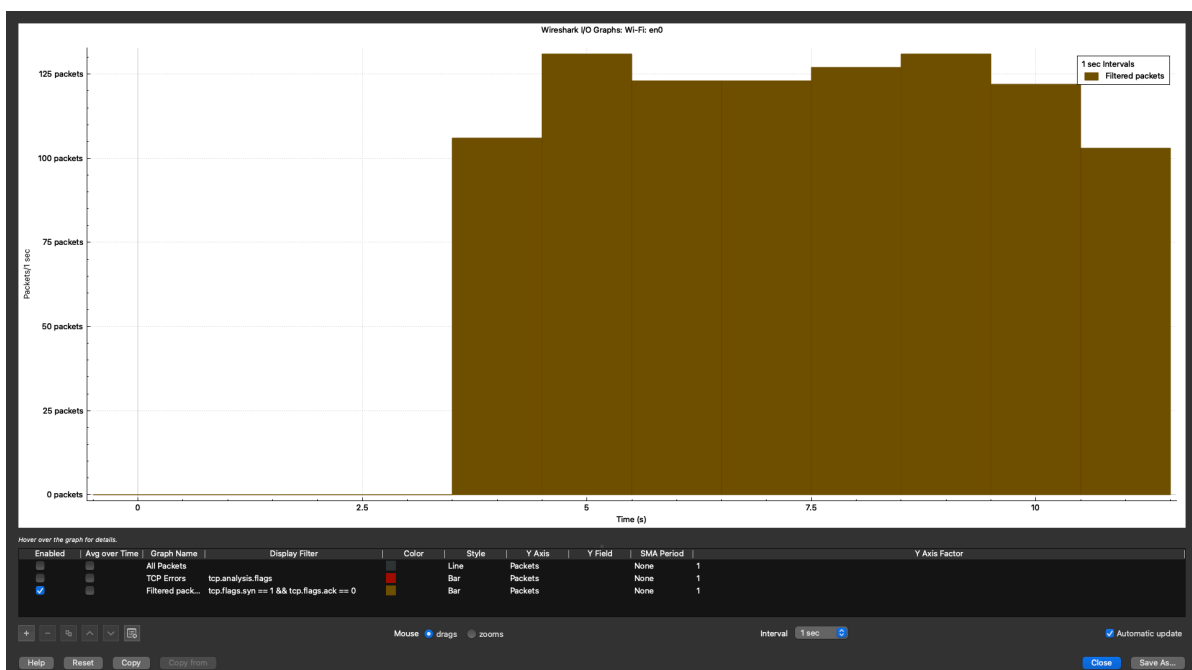
- Apply this filter: `tcp.flags.syn == 1 && tcp.flags.ack == 0`



The image shows a Wireshark packet capture window with the filter `tcp.flags.syn == 1 && tcp.flags.ack == 0` applied. The packet list displays a series of TCP SYN packets from various source IP addresses to the destination 192.168.1.57. The packet details pane shows the structure of a TCP segment, including the header and options. The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
4344	4.225863	63.18.182.41	192.168.1.57	TCP	54	46481 → 80 [SYN] Seq=0 Win=8192 Len=0
4347	4.229488	63.18.182.41	192.168.1.57	TCP	58	[TCP Retransmission] 46481 → 80 [SYN] Seq=0 Win=8192 Len=0
4369	4.238586	215.146.185.186	192.168.1.57	TCP	54	68157 → 80 [SYN] Seq=0 Win=8192 Len=0
4370	4.248101	215.146.185.186	192.168.1.57	TCP	58	[TCP Retransmission] 68157 → 80 [SYN] Seq=0 Win=8192 Len=0
4372	4.251727	2.227.228.172	192.168.1.57	TCP	54	53657 → 80 [SYN] Seq=0 Win=8192 Len=0
4377	4.255548	2.227.228.172	192.168.1.57	TCP	58	[TCP Retransmission] 53657 → 80 [SYN] Seq=0 Win=8192 Len=0
4387	4.264795	194.62.98.191	192.168.1.57	TCP	54	65239 → 80 [SYN] Seq=0 Win=8192 Len=0
4388	4.264795	194.62.98.191	192.168.1.57	TCP	58	[TCP Retransmission] 65239 → 80 [SYN] Seq=0 Win=8192 Len=0
4418	4.277828	62.183.98.6	192.168.1.57	TCP	54	43648 → 80 [SYN] Seq=0 Win=8192 Len=0
4411	4.281728	62.183.98.6	192.168.1.57	TCP	58	[TCP Retransmission] 43648 → 80 [SYN] Seq=0 Win=8192 Len=0
4417	4.298862	171.38.198.223	192.168.1.57	TCP	54	15678 → 80 [SYN] Seq=0 Win=8192 Len=0
4420	4.303318	171.38.198.223	192.168.1.57	TCP	58	[TCP Retransmission] 15678 → 80 [SYN] Seq=0 Win=8192 Len=0
4448	4.383955	33.238.49.197	192.168.1.57	TCP	54	35298 → 80 [SYN] Seq=0 Win=8192 Len=0
4447	4.389338	33.238.49.197	192.168.1.57	TCP	58	[TCP Retransmission] 35298 → 80 [SYN] Seq=0 Win=8192 Len=0
4455	4.317818	38.242.97.207	192.168.1.57	TCP	54	65278 → 80 [SYN] Seq=0 Win=8192 Len=0
4456	4.320405	38.242.97.207	192.168.1.57	TCP	58	[TCP Retransmission] 65278 → 80 [SYN] Seq=0 Win=8192 Len=0
4478	4.338439	16.35.127.18	192.168.1.57	TCP	54	1347 → 80 [SYN] Seq=0 Win=8192 Len=0
4471	4.333756	16.35.127.18	192.168.1.57	TCP	58	[TCP Retransmission] 1347 → 80 [SYN] Seq=0 Win=8192 Len=0
4489	4.343795	185.158.47.183	192.168.1.57	TCP	54	58887 → 80 [SYN] Seq=0 Win=8192 Len=0
4490	4.348081	185.158.47.183	192.168.1.57	TCP	58	[TCP Retransmission] 58887 → 80 [SYN] Seq=0 Win=8192 Len=0
4584	4.355181	55.29.251.186	192.168.1.57	TCP	54	4713 → 80 [SYN] Seq=0 Win=8192 Len=0
4585	4.358174	55.29.251.186	192.168.1.57	TCP	58	[TCP Retransmission] 4713 → 80 [SYN] Seq=0 Win=8192 Len=0
4515	4.368438	72.184.77.144	192.168.1.57	TCP	54	23338 → 80 [SYN] Seq=0 Win=8192 Len=0
4516	4.371743	72.184.77.144	192.168.1.57	TCP	58	[TCP Retransmission] 23338 → 80 [SYN] Seq=0 Win=8192 Len=0
4534	4.318876	216.167.89.185	192.168.1.57	TCP	54	48529 → 80 [SYN] Seq=0 Win=8192 Len=0
4539	4.382336	216.167.89.185	192.168.1.57	TCP	58	[TCP Retransmission] 48529 → 80 [SYN] Seq=0 Win=8192 Len=0
4553	4.393518	198.71.26.166	192.168.1.57	TCP	54	44933 → 80 [SYN] Seq=0 Win=8192 Len=0
4554	4.398259	198.71.26.166	192.168.1.57	TCP	58	[TCP Retransmission] 44933 → 80 [SYN] Seq=0 Win=8192 Len=0
4564	4.407128	211.141.44.188	192.168.1.57	TCP	54	1470 → 80 [SYN] Seq=0 Win=8192 Len=0
4569	4.418873	211.141.44.188	192.168.1.57	TCP	58	[TCP Retransmission] 1470 → 80 [SYN] Seq=0 Win=8192 Len=0
4589	4.422268	139.161.212.379	192.168.1.57	TCP	54	6866 → 80 [SYN] Seq=0 Win=8192 Len=0
4598	4.425358	19.121.132.179	192.168.1.57	TCP	58	[TCP Retransmission] 6866 → 80 [SYN] Seq=0 Win=8192 Len=0
4608	4.437768	239.163.225.280	192.168.1.57	TCP	54	45311 → 80 [SYN] Seq=0 Win=8192 Len=0
4628	4.452559	177.244.151.36	192.168.1.57	TCP	54	27623 → 80 [SYN] Seq=0 Win=8192 Len=0
4630	4.455959	177.244.151.36	192.168.1.57	TCP	58	[TCP Retransmission] 27623 → 80 [SYN] Seq=0 Win=8192 Len=0
4652	4.468471	171.18.158.74	192.168.1.57	TCP	54	32593 → 80 [SYN] Seq=0 Win=8192 Len=0
4653	4.474332	171.18.158.74	192.168.1.57	TCP	58	[TCP Retransmission] 32593 → 80 [SYN] Seq=0 Win=8192 Len=0
4663	4.483191	67.137.43.177	192.168.1.57	TCP	54	11929 → 80 [SYN] Seq=0 Win=8192 Len=0
4668	4.486715	67.137.43.177	192.168.1.57	TCP	58	[TCP Retransmission] 11929 → 80 [SYN] Seq=0 Win=8192 Len=0
4694	4.499646	235.48.78.237	192.168.1.57	TCP	54	43675 → 80 [SYN] Seq=0 Win=8192 Len=0
4785	4.515483	13.140.134.118	192.168.1.57	TCP	54	37424 → 80 [SYN] Seq=0 Win=8192 Len=0
4714	4.518482	13.140.134.118	192.168.1.57	TCP	58	[TCP Retransmission] 37424 → 80 [SYN] Seq=0 Win=8192 Len=0
4738	4.531619	182.254.178.198	192.168.1.57	TCP	54	21265 → 80 [SYN] Seq=0 Win=8192 Len=0
4745	4.535616	182.254.178.198	192.168.1.57	TCP	58	[TCP Retransmission] 21265 → 80 [SYN] Seq=0 Win=8192 Len=0

- Normal TCP handshake: SYN→SYN-ACK→ACK
In your flood simulation: SYN→SYN→SYN
- More than 100 packets in 1 sec.



- SYN packets are small around 60 bytes

Topic / Item	Count	Average	Min Value	Max Value	Rate (ms)	Percent	Burst Rate	Burst Start
Packet Lengths	966	56.05	54	98	0.1274	100%	0.2200	5.750
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	963	55.92	54	58	0.1270	99.69%	0.2200	5.750
80-159	3	98.00	98	98	0.0004	0.31%	0.0100	9.334
160-319	0	-	-	-	0.0000	0.00%	-	-
320-639	0	-	-	-	0.0000	0.00%	-	-
640-1279	0	-	-	-	0.0000	0.00%	-	-
1280-2559	0	-	-	-	0.0000	0.00%	-	-
2560-5119	0	-	-	-	0.0000	0.00%	-	-
5120 and greater	0	-	-	-	0.0000	0.00%	-	-

Q2. HTTP Flood Attack

The packet capture was analyzed using Wireshark. Filtering HTTP GET requests revealed a large number of repeated requests generated by the simulation script. The IO graph showed a significant spike in packet rate during the test period, indicating abnormal traffic behavior. Endpoint statistics confirmed that a single host generated the majority of the requests toward the target server. The rapid timing between packets and repeated request patterns demonstrate characteristics of an HTTP flood attack.

The image displays a Wireshark packet capture analysis. The top pane shows a list of captured packets, all of which are HTTP GET requests from the source IP 192.168.1.57 to the destination IP 192.168.1.55. The middle pane shows the details of a selected packet, which is an HTTP GET request. The bottom pane shows the raw packet data in hexadecimal and ASCII format.

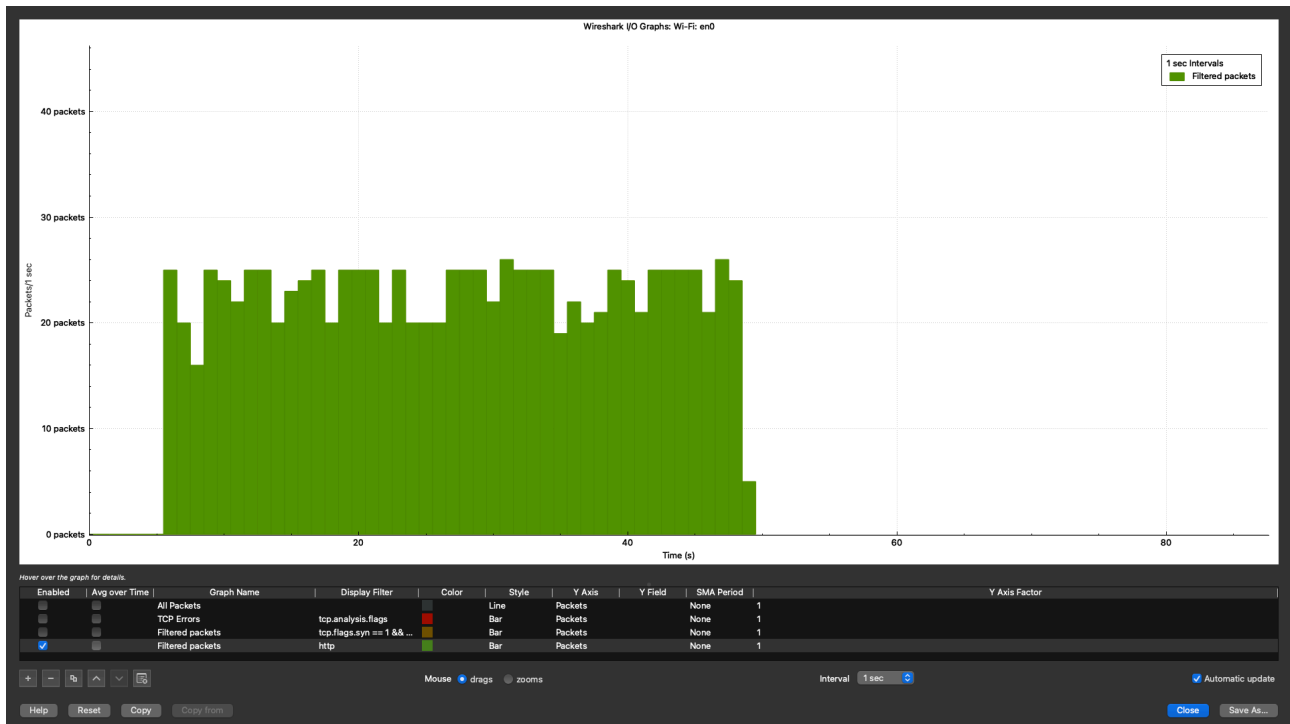
No.	Time	Source	Destination	Protocol	Length	Info
79	6.816512	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
80	6.820146	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
84	6.825598	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
88	6.831798	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
92	6.841190	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
107	6.216954	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
109	6.223986	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
110	6.235617	192.168.1.57	192.168.1.55	HTTP	1189	HTTP/1.1 200 OK (text/html)
111	6.245580	192.168.1.57	192.168.1.55	HTTP	1189	HTTP/1.1 200 OK (text/html)
112	6.252142	192.168.1.57	192.168.1.55	HTTP	1189	HTTP/1.1 200 OK (text/html)
119	6.429813	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
126	6.441871	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
132	6.447446	192.168.1.57	192.168.1.55	HTTP	544	HTTP/1.1 200 OK (text/html)
136	6.463539	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
137	6.466119	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
165	6.647782	192.168.1.57	192.168.1.55	HTTP	1189	HTTP/1.1 200 OK (text/html)
168	6.665558	192.168.1.57	192.168.1.55	HTTP	1189	HTTP/1.1 200 OK (text/html)
169	6.669543	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
170	6.672553	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
200	6.862380	192.168.1.57	192.168.1.55	HTTP	1189	HTTP/1.1 200 OK (text/html)
201	6.865962	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
205	6.872529	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
212	6.880634	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
213	6.884158	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
224	7.145809	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
226	7.150873	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
232	7.156392	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
234	7.161185	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
244	7.180464	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
251	7.397187	192.168.1.57	192.168.1.55	HTTP	1189	HTTP/1.1 200 OK (text/html)
254	7.410213	192.168.1.57	192.168.1.55	HTTP	144	HTTP/1.1 200 OK (text/html)
258	7.416779	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
262	7.423143	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
263	7.425600	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
276	7.617856	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
278	7.639275	192.168.1.57	192.168.1.55	HTTP	1189	HTTP/1.1 200 OK (text/html)
282	7.654748	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
286	7.661285	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
290	7.678373	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
324	7.924632	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
328	7.931952	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
332	7.955972	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
336	7.966398	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)

Frame 2695: Packet, 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface en0, id 0
Ethernet II, Src: A6:4C:8C:39:EE:25 (A6:4C:8C:39:EE:25), Dst: A6:4C:8C:39:EE:25 (A6:4C:8C:39:EE:25)
Internet Protocol Version 4, Src: 192.168.1.57, Dst: 192.168.1.55
Transmission Control Protocol, Src Port: 80, Dst Port: 52618, Seq: 37578, Ack: 7426, Len: 5
[Reassembled TCP Segments (1155 bytes): #2692(653), #2693(424), #2694(73), #2695(5)]
Hypertext Transfer Protocol, has 4 chunks (including last chunk)
Line-based text data: text/html (29 lines)

Packet 2695: Reassembled TCP (1...), De-chunked entity body L...

Packets: 4010 - Displayed: 1000 (24.9%) - Dropped: 0 (0.0%) Profile: Default

- Flood simulate more than 2-5 packets per second



- Time between packets is very less

No.	Time	Source	Destination	Protocol	Length	Info
79	6.016512	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
80	6.020140	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
84	6.025598	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
88	6.031790	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)
92	6.041190	192.168.1.57	192.168.1.55	HTTP	71	HTTP/1.1 200 OK (text/html)

- Packet size pattern

Topic / Item	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
Packet Lengths	1000	362.56	71	1190	0.0232	100%	0.0600	16.179
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	730	71.00	71	71	0.0169	73.00%	0.0600	30.507
80-159	5	144.00	144	144	0.0001	0.50%	0.0100	6.447
160-319	0	-	-	-	0.0000	0.00%	-	-
320-639	8	562.00	562	562	0.0002	0.80%	0.0100	8.425
640-1279	257	1188.77	1152	1190	0.0060	25.70%	0.0500	27.925
1280-2559	0	-	-	-	0.0000	0.00%	-	-
2560-5119	0	-	-	-	0.0000	0.00%	-	-
5120 and greater	0	-	-	-	0.0000	0.00%	-	-